



SAIR

Spatial AI & Robotics Lab

CSE 473/573

L18: RECOGNITION

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Objects vs Texture vs Scene

The texture



The object



The scene



Scenes vs. objects

A photograph of a firehydrant



A photograph of a street



Scene views vs. objects

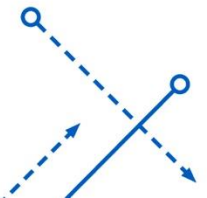
- Scene:
 - a place in which a human can act within
 - or a place to which a human being could navigate.



- A lot more than just a combination of objects
 - just as objects are more than the combinations of their parts.
- Associated with specific functions and behaviors (like objects)
 - E.g., eating in a restaurant, reading in a library,
 - Talking in classroom, playing in a park, etc

Why be concerned about the difference?

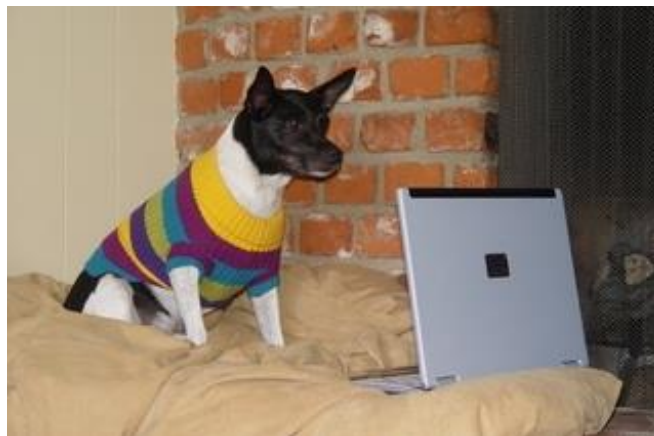
- Features to be extracted?
- Context for Recognition?
- Various **categories** have features that can be essential for interpretation.



Why do we care about categories?

Perception of function:

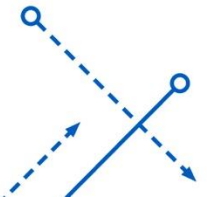
- We can perceive the 3D shape, texture, material properties, without knowing about objects.
- But, the concept of category also encodes information about what can we do with those objects.



“We therefore include the perception of function as a proper –indeed, crucial- subject for vision science”, *from Vision Science, chapter 9, Palmer.*

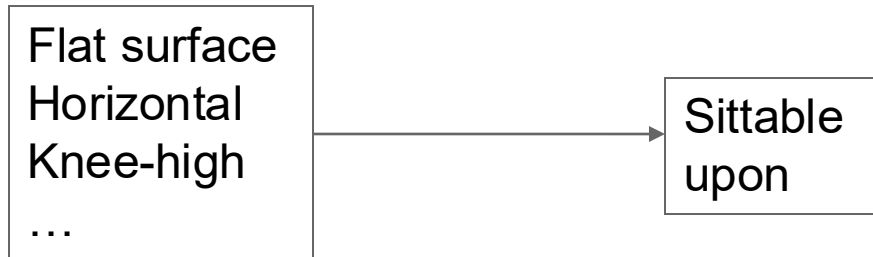
Why do we care about categories?

- When we recognize an object, we can make
 - predictions about its behavior in the future
 - beyond of what is immediately perceived.

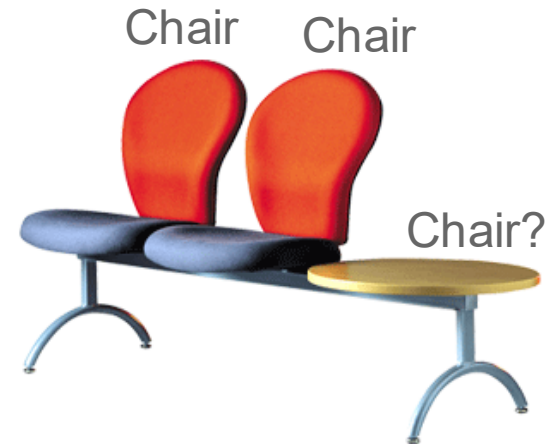
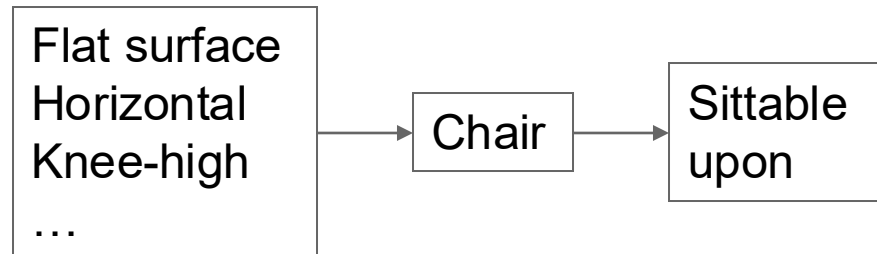


The perception of function

- Direct perception (affordances): [introduced by Gibson]



- Mediated perception (Categorization)



- One caveat of this comparison:
 - Deciding that something is a chair might require access to more features than deciding that we can sit on something
 - A different level of categorization.

Direct perception

Some aspects of an object function can be perceived directly

- Functional form:
 - Some forms clearly indicate to a function
 - e.g., “sittable-upon”, container, cutting device, ...

Sittable-upon

Sittable-upon



It does not seem easy to sit-upon this...



Limitations of Direct Perception

- Objects of similar structure might have very different functions



Figure 9.1.2 Objects with similar structure but different functions. Mailboxes afford letter mailing, whereas trash cans do not, even though they have many similar physical features, such as size, location, and presence of an opening large enough to insert letters and medium-sized packages.



Not all functions seem to be available from direct visual information only.

- The functions are the same at some level of description:
 - We can put things inside in both
 - Somebody will come later to empty them.
- However, we don't put inside the same kinds of things

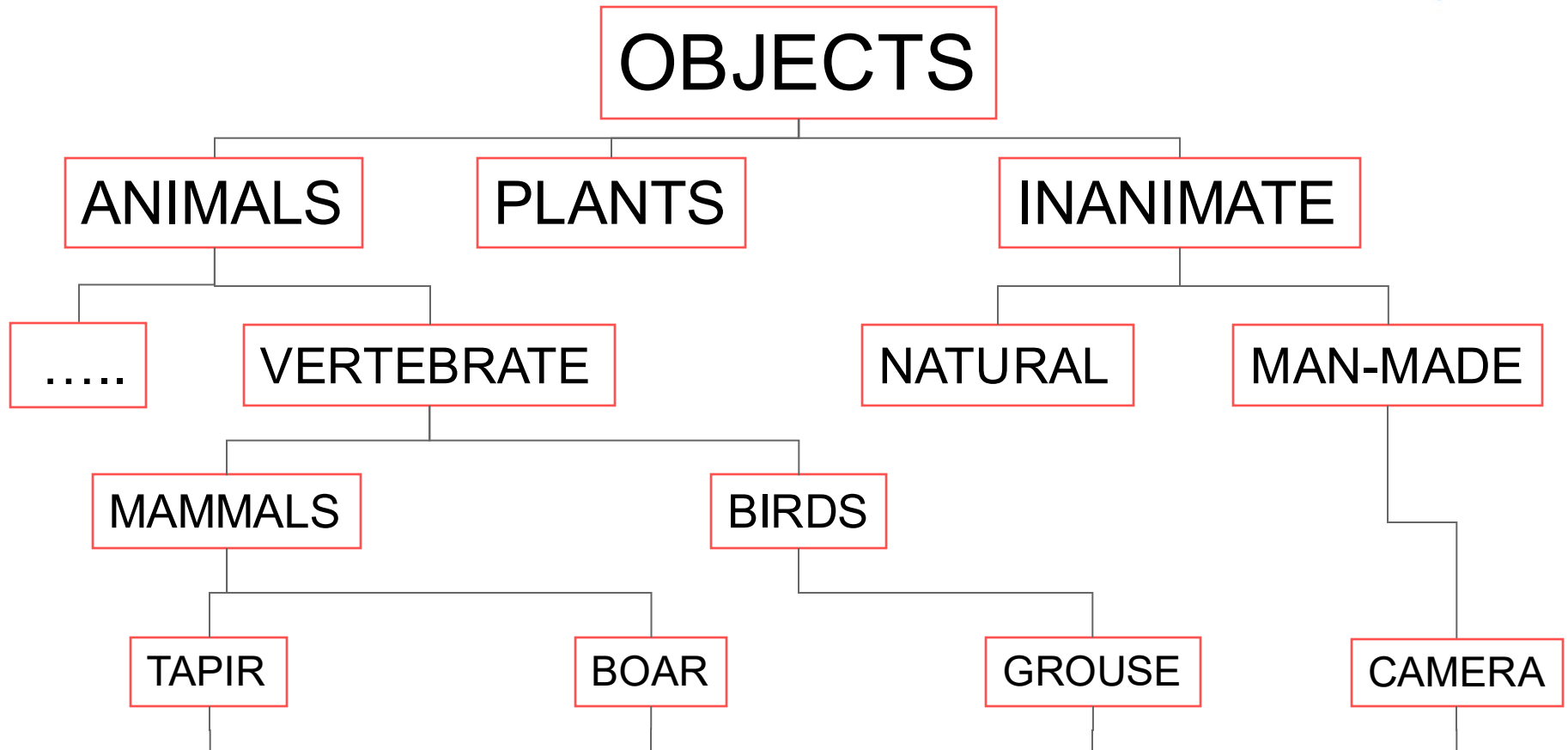
Which level of categorization is the right one?

- Car is an object composed of:
 - a few doors, four wheels (not always visible), a roof
 - front lights, windshield



If you are thinking in buying a car, you might want to be a bit more specific about your categorization.

Object Category



Entry-level categories

- Typical member of a basic-level category are categorized at the expected level
- Atypical members tend to be classified at a finer level.



A bird

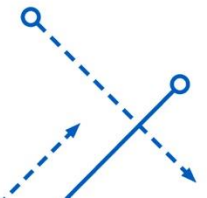


An ostrich

Demo : Rapid image understanding

Instructions: 9 photographs will be shown for half a second each. Your task is to **memorize these pictures.**

GET READY



Memory Test



Have you seen this picture ?

Memory Test



Memory Test



Have you seen this picture ?

Memory Test



Memory Test



Have you seen this picture ?

Memory Test



Memory Test



Have you seen this picture ?

Memory Test



Memory Test



Have you seen this picture ?

Memory Test



Memory Test



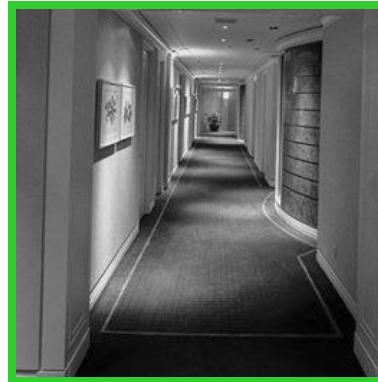
Have you seen this picture ?

Memory Test

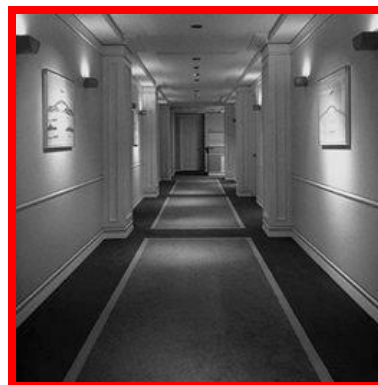


Memory Test Overview

You have seen these pictures



You were tested with these pictures



The gist of the scene

In a glance, we remember the meaning of an image and its global **layout** but some objects and **details** are forgotten



Mary Potter (1976)

Mary Potter (1975, 1976) demonstrated

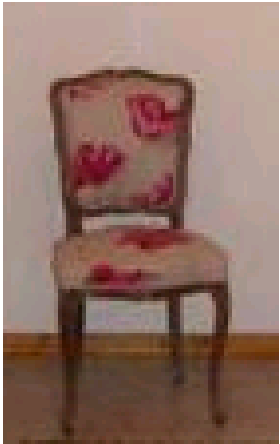
- during a rapid sequential visual presentation (100 msec per image), a novel picture is instantly **understood** and observers seem to comprehend a lot of visual information



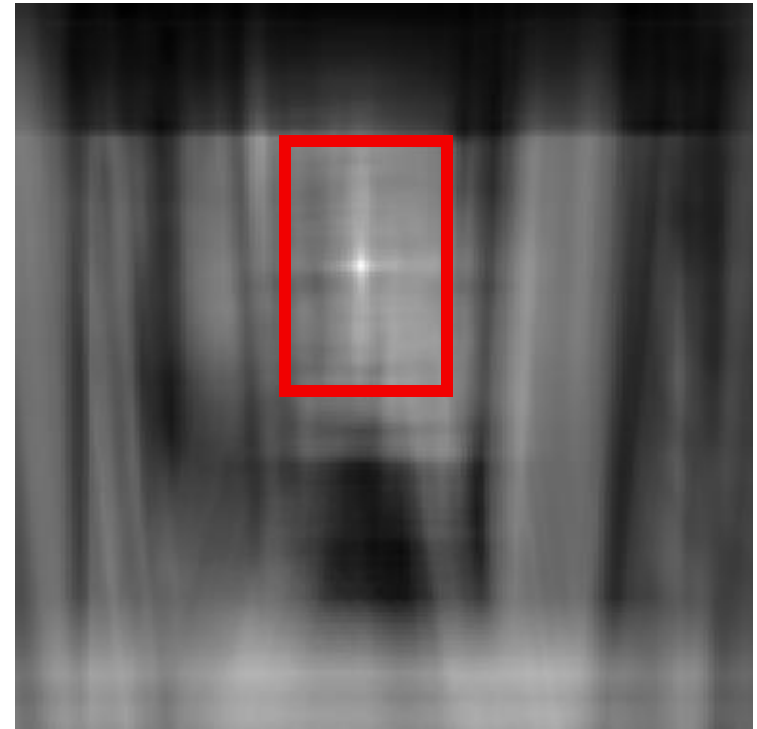
Object recognition: Is it really so hard?

Find the chair in this image

This is a chair

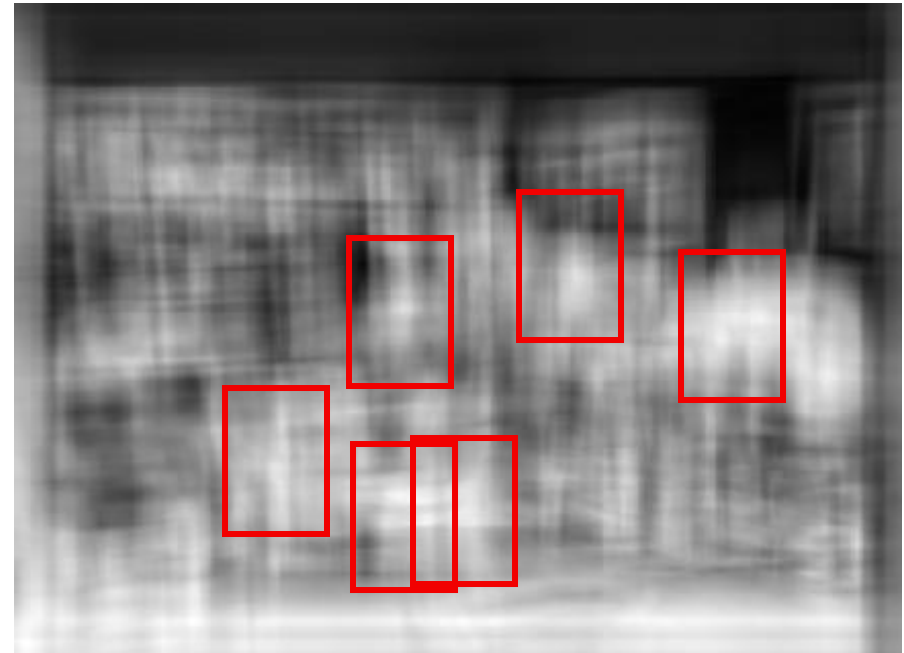
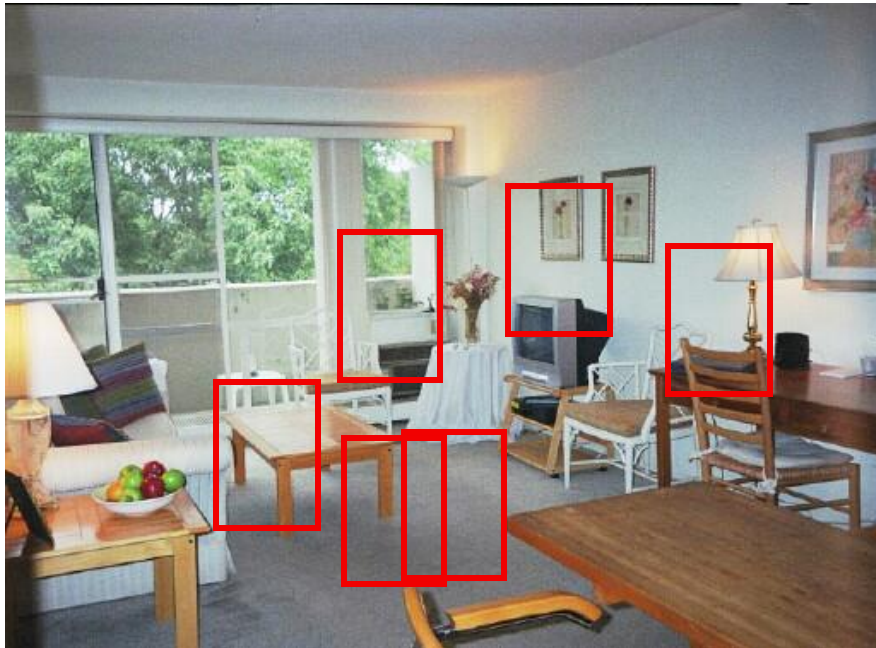
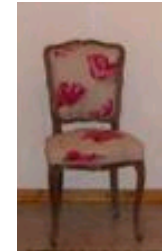


Output of normalized correlation



Object recognition: Is it really so hard?

Find the chair in this image



Pretty much garbage
Template matching cannot make it.

Challenges: robustness



Illumination



Object pose



Clutter



Occlusions



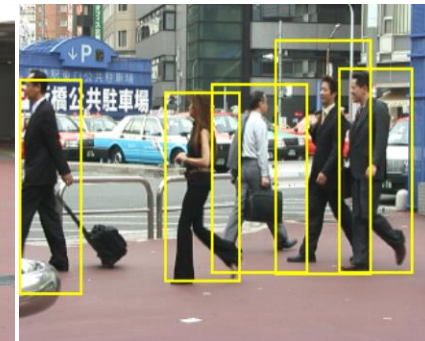
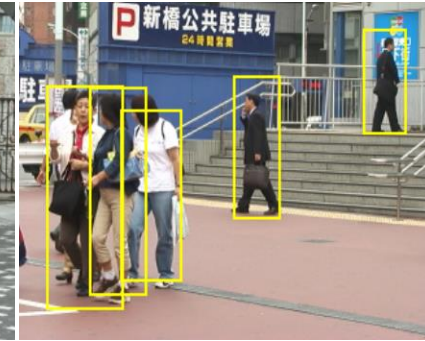
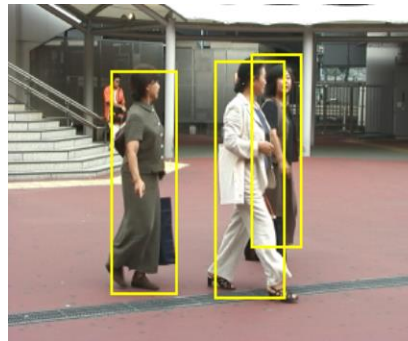
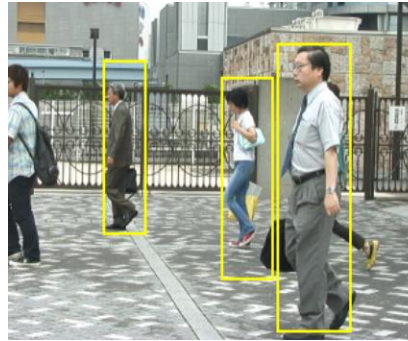
**Intra-class
appearance**



Viewpoint

Challenges: robustness/invariance

- Detection in Crowded Scenes
 - Learn object variability
 - Changes in appearance, scale, and articulation
 - Compensate for clutter, overlap, and occlusion



Challenges: context and human experience

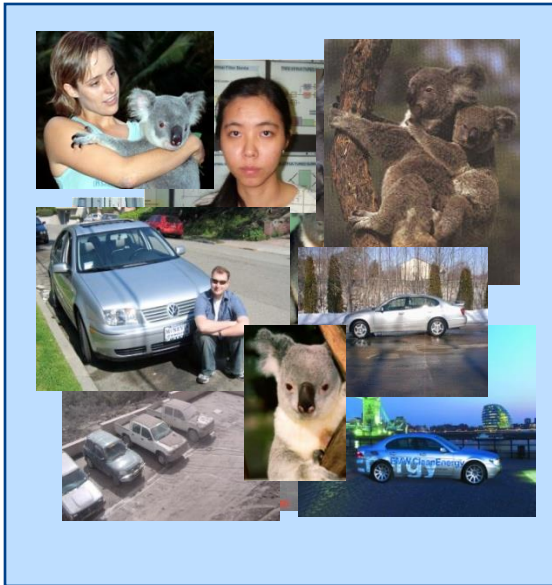


Challenges: learning with minimal supervision

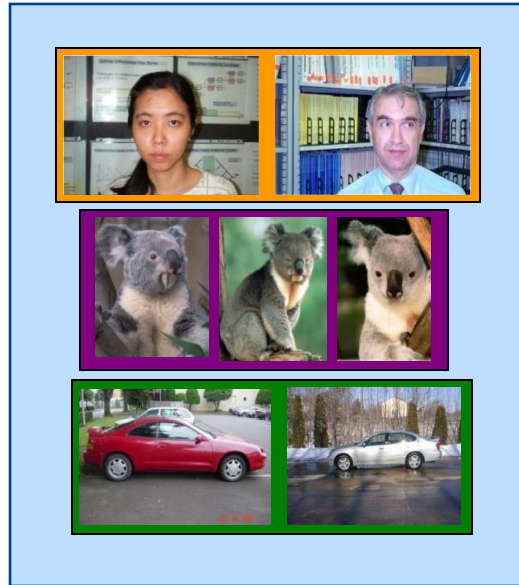
Less
!

!

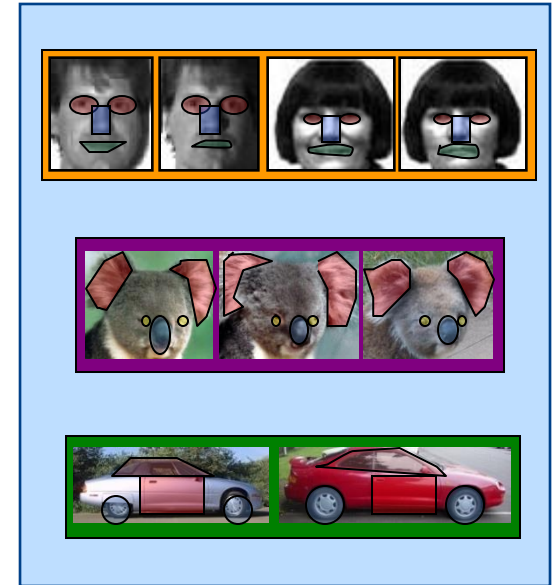
More
!



Unlabeled,
multiple objects



Classes labeled,
some clutter



Cropped to object,
parts and classes
labeled

Inputs/outputs/assumptions

What is the **goal**?

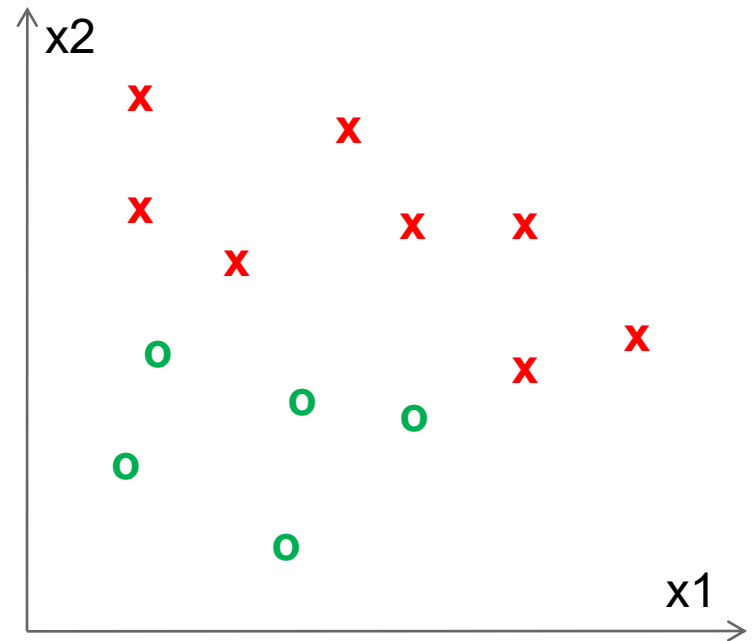
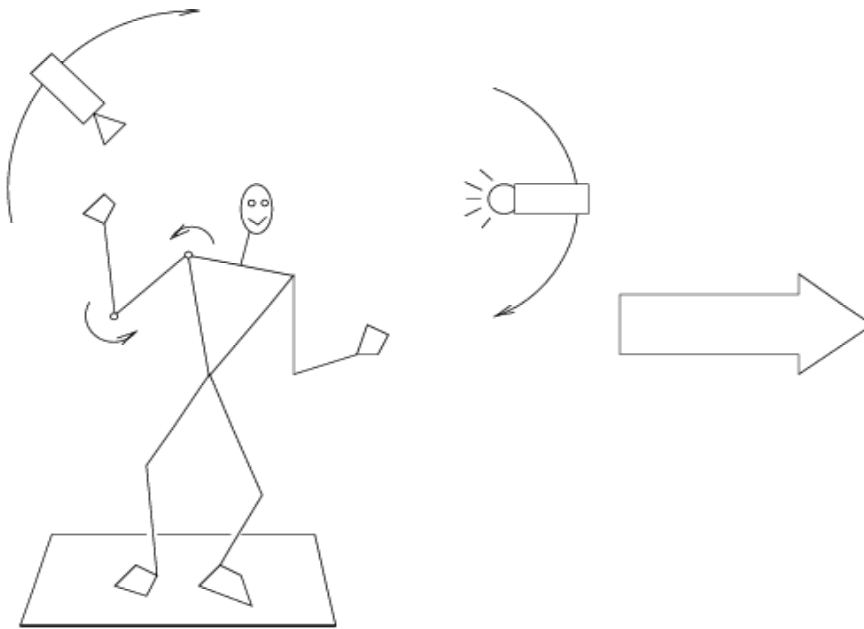
- Say yes/no as to whether an object present in image

And/or:

- Categorize all objects
- Forced choice from pool of categories
- Bounding box on object
- Full segmentation
- Build a model of an object category
- Determine pose of an object, e.g., for robot to grasp

Recognition is all about modeling variability

- Invariance is the key challenge to computer vision.



Variability:

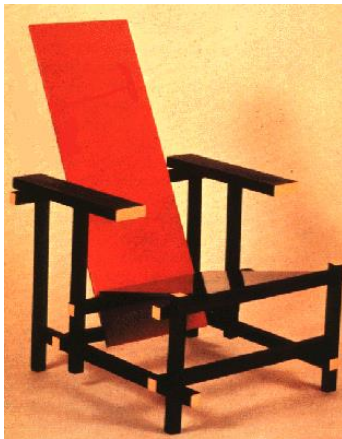
Camera position

Illumination

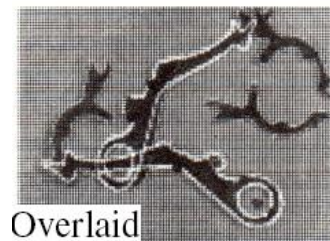
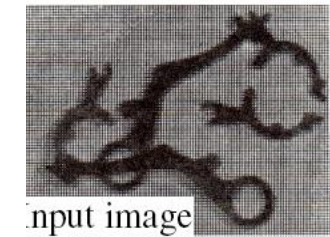
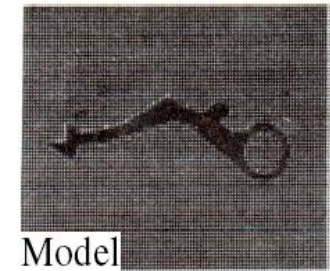
Shape parameters

Within-class variations?

Within-class variations



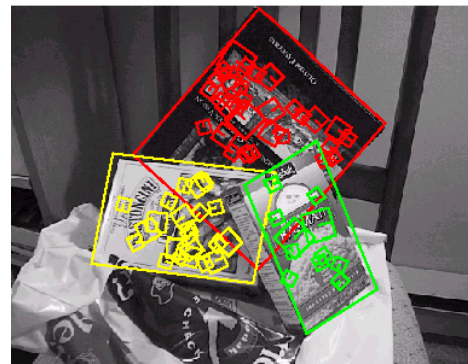
Rough Evolution of Focus in Object Recognition



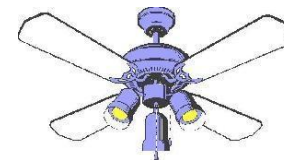
1980s



7 5 9 2 6 5
2 2 2 2 2 3
0 2 3 8 0 7

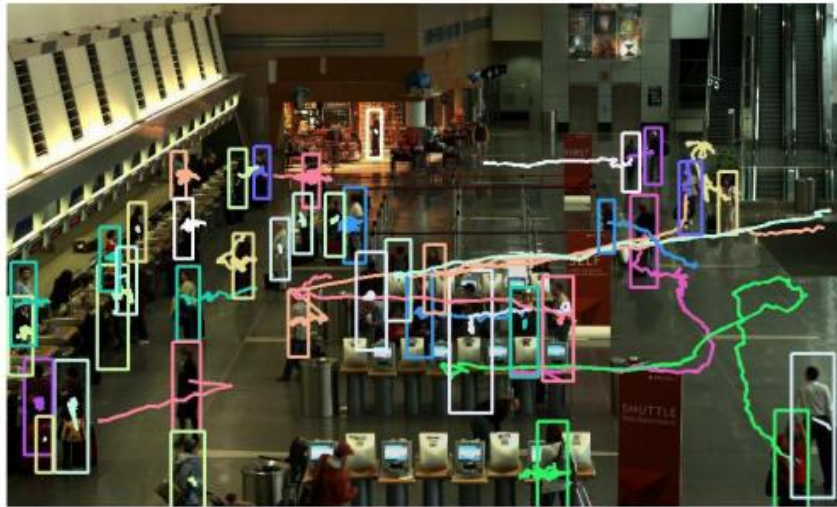


1990s to early 2000s



2000-2010...

Today



History of ideas in recognition

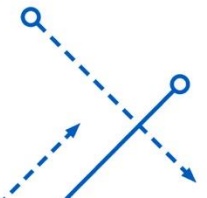
- 1960s – early 1990s: the geometric era

Variability: Camera position
Illumination

Focus: **Alignment**

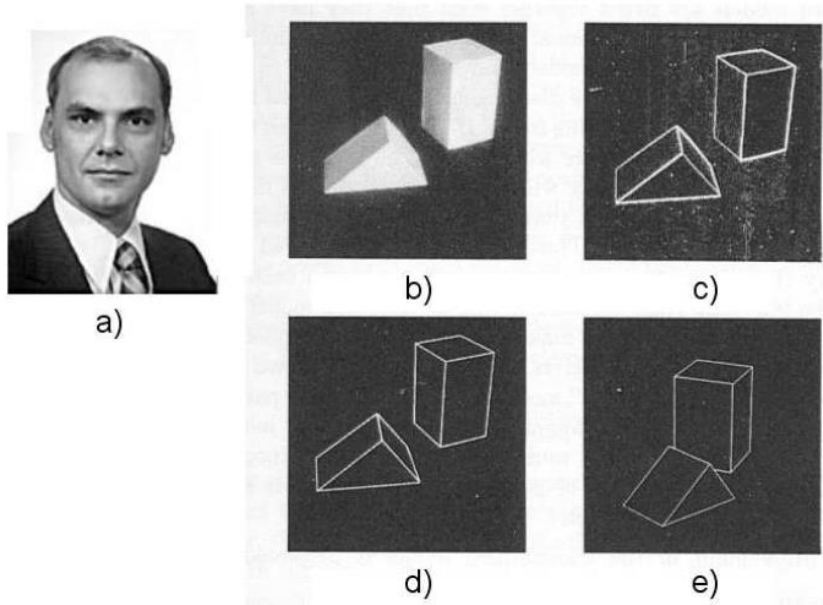
Shape: Assumed known

Roberts (1965); Lowe (1987); Faugeras & Hebert (1986);
Grimson & Lozano-Perez (1986); Huttenlocher & Ullman (1987)



History of ideas in recognition

- Recognition as an alignment problem: Block world



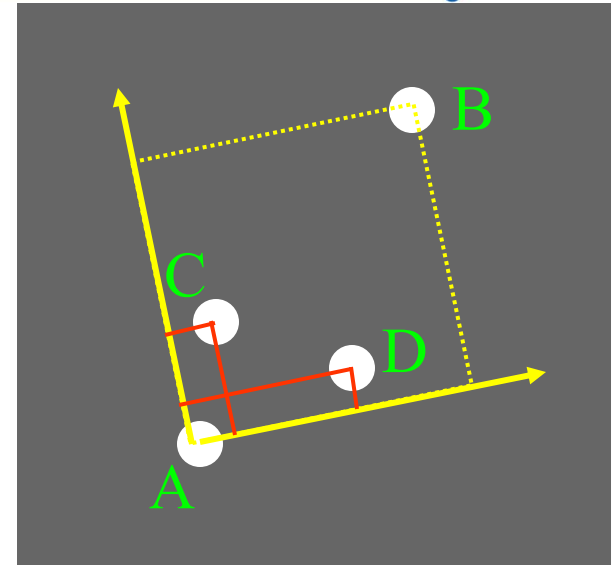
L. G. Roberts, [*Machine Perception of Three Dimensional Solids*](#), Ph.D. thesis, MIT Department of Electrical Engineering, 1963.

Fig. 1. A system for recognizing 3-d polyhedral scenes. a) L.G. Roberts. b) A blocks world scene. c) Detected edges using a 2x2 gradient operator. d) A 3-d polyhedral description of the scene, formed automatically from the single image. e) The 3-d scene displayed with a viewpoint different from the original image to demonstrate its accuracy and completeness. (b) - e) are taken from [64] with permission MIT Press.)

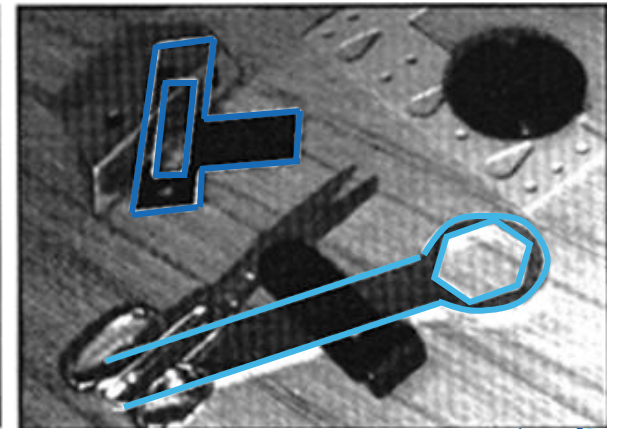
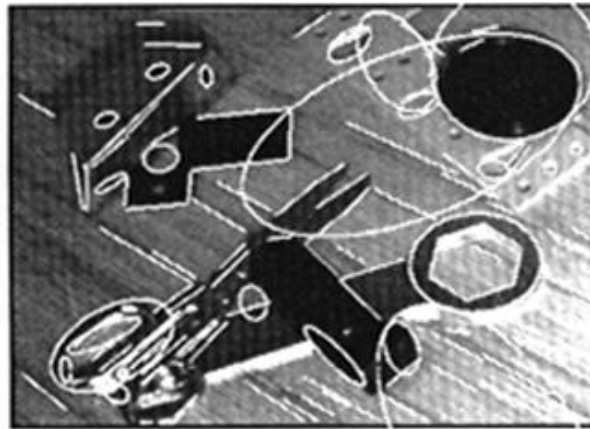
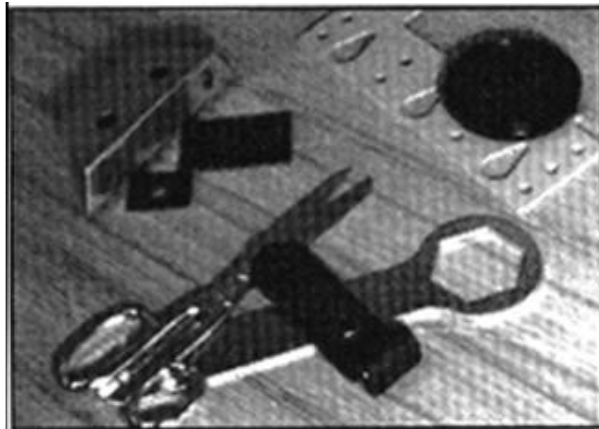
History of ideas in recognition

Example:
invariant to similarity transformations
computed from four points

General 3D objects do not admit monocular viewpoint
invariants (Burns et al., 1993)



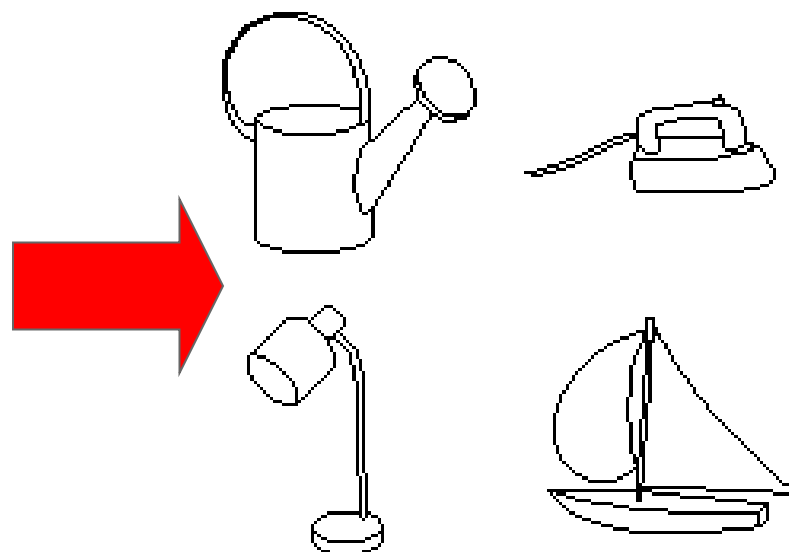
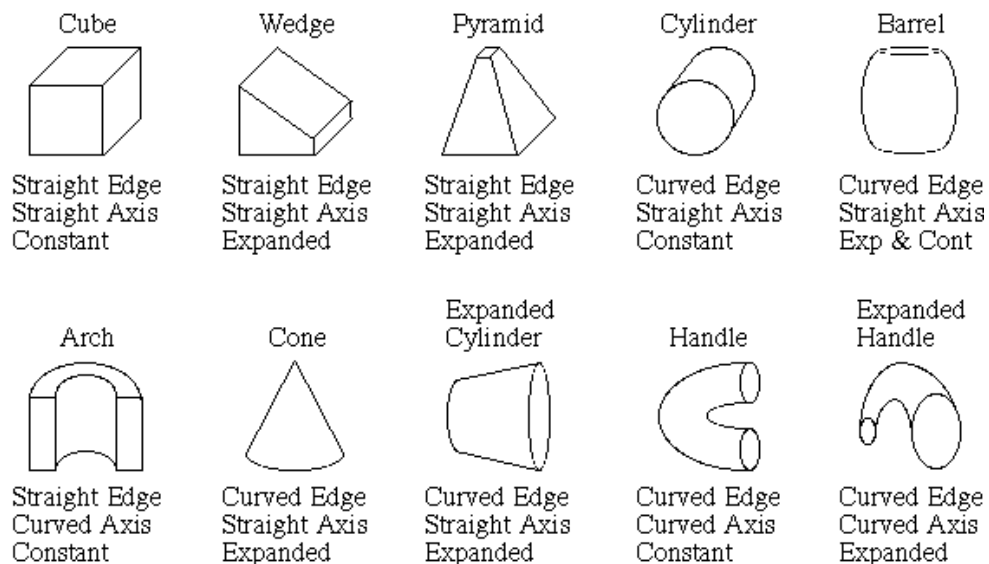
Projective invariants (Rothwell et al., 1992):

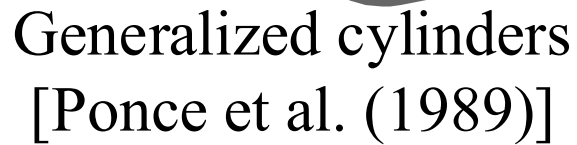


Recognition by components

Primitives (geons) Biederman (1987)

Objects





History of ideas in recognition

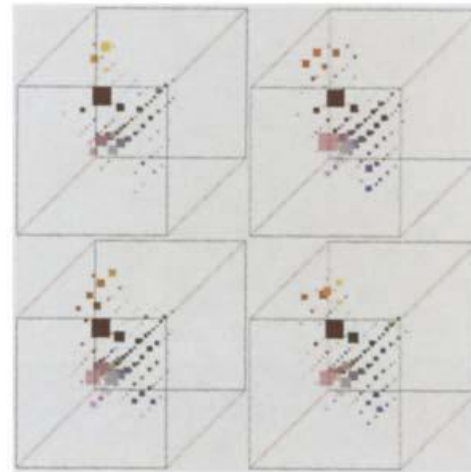
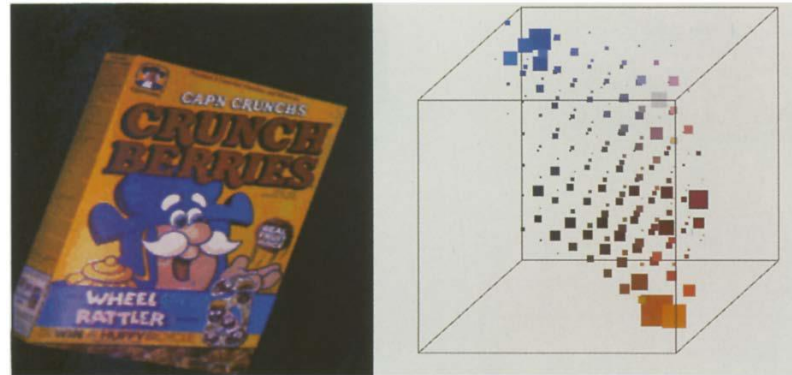
- 1960s – early 1990s: the geometric era
- 1990s: appearance-based models

Empirical models of image variability

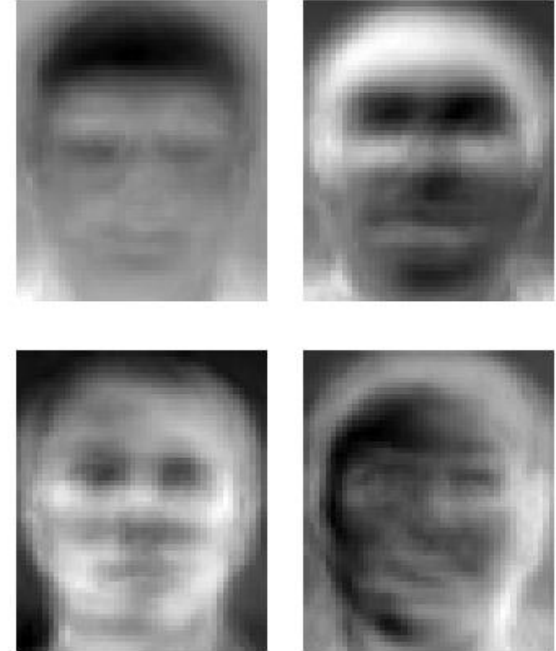
Appearance-based techniques

Turk & Pentland (1991); Murase & Nayar (1995); etc.

Color Histograms



Eigenfaces (Turk & Pentland, 1991)



Experimental Condition	Correct/Unknown Recognition Percentage		
	Lighting	Orientation	Scale
Forced classification	96/0	85/0	64/0
Forced 100% accuracy	100/19	100/39	100/60
Forced 20% unknown rate	100/20	94/20	74/20

History of ideas in recognition

- 1960s – early 1990s: the geometric era
- 1990s: appearance-based models
- Mid-1990s: sliding window approaches



Sliding window approaches



Turk and Pentland, 1991
Belhumeur, Hespanha, & Kriegman, 1997
Schneiderman & Kanade 2004
Viola and Jones, 2000

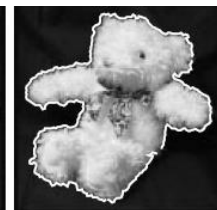


Schneiderman & Kanade, 2004
Argawal and Roth, 2002
Poggio et al. 1993

History of ideas in recognition

- 1960s – early 1990s: the geometric era
- 1990s: appearance-based models
- Mid-1990s: sliding window approaches
- Late 1990s: local features

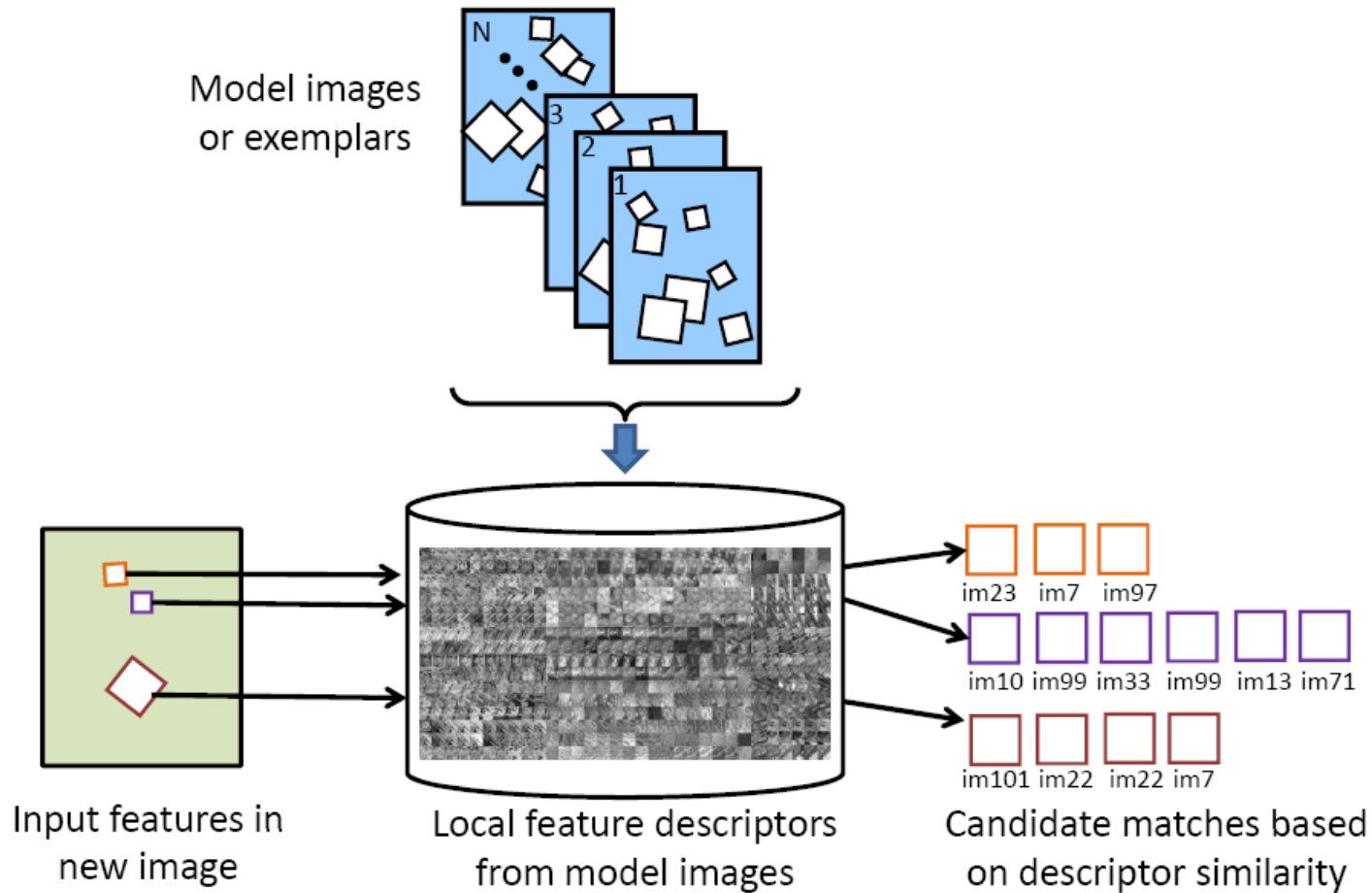
Local features for instance recognition



D. Lowe (1999, 2004)

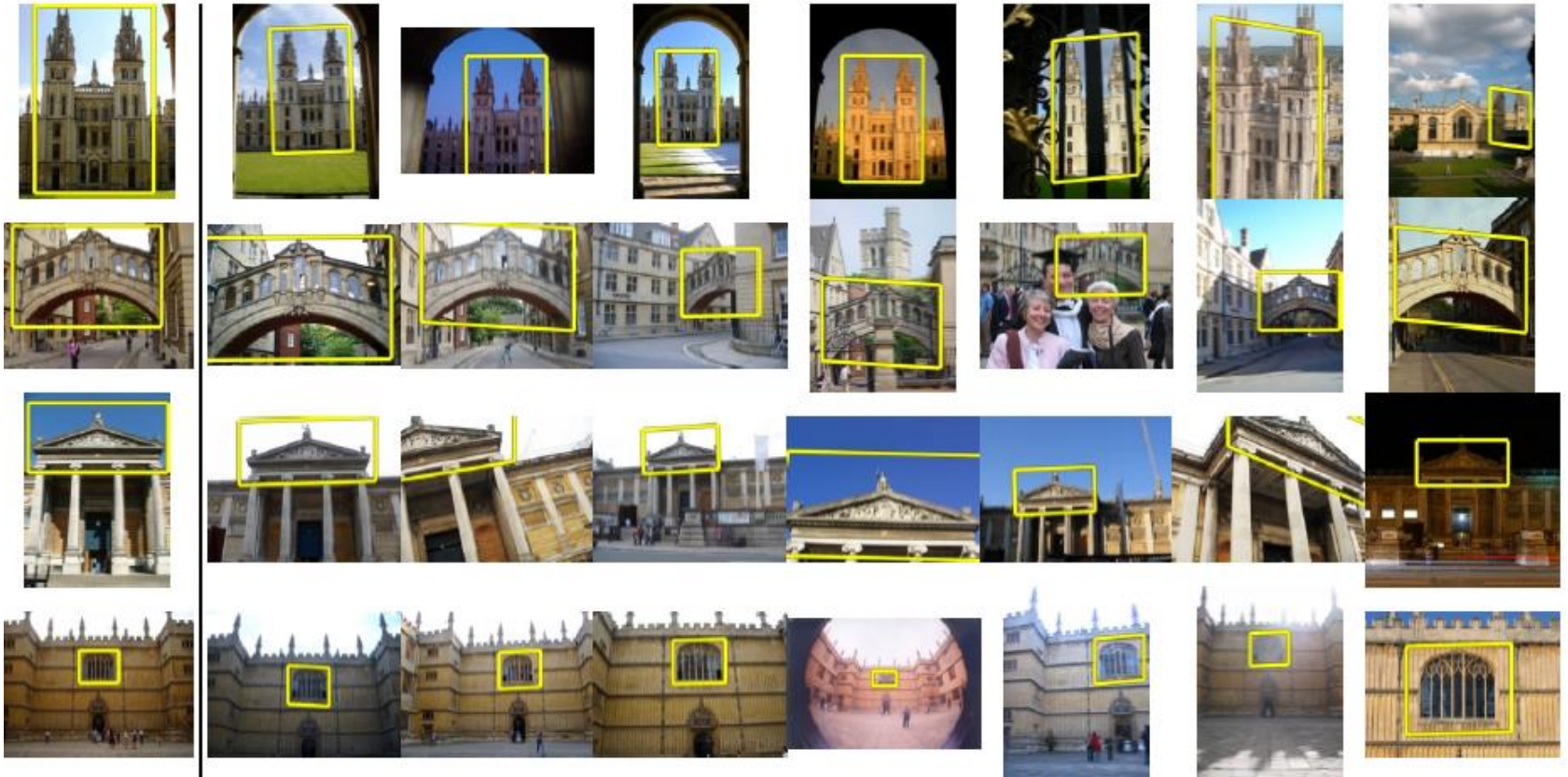
Large-scale image search

Combining local features, indexing, and spatial constraints



Large-scale image search

Combining local features, indexing, and spatial constraints



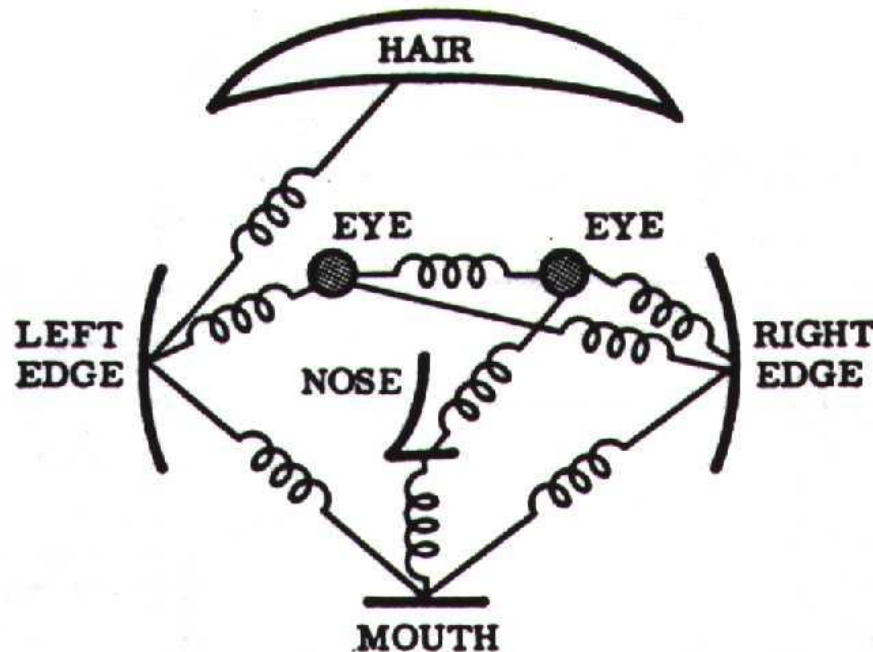
Philbin et al. '07

History of ideas in recognition

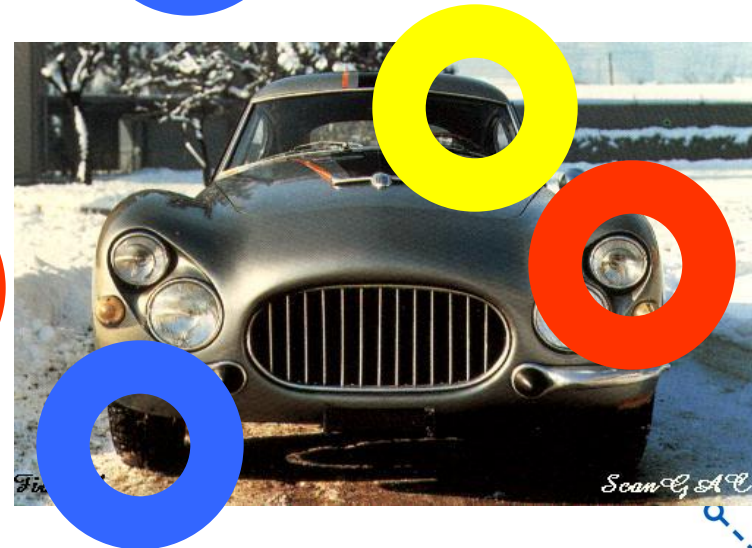
- 1960s – early 1990s: the geometric era
- 1990s: appearance-based models
- Mid-1990s: sliding window approaches
- Late 1990s: local features
- Early 2000s: parts-and-shape models

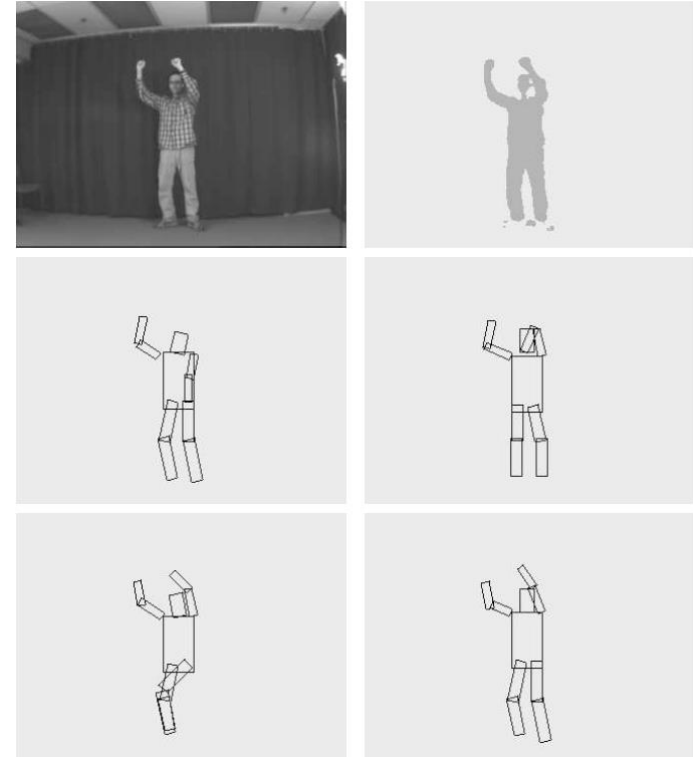
Parts-and-shape models

- Model:
 - Object as a set of parts
 - **Relative locations** between parts
 - **Appearance of part**

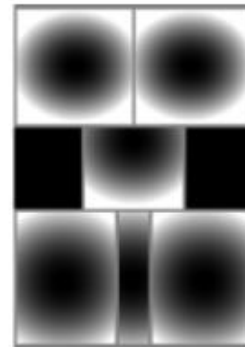
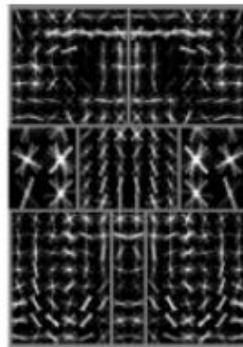
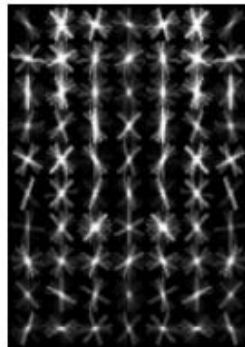
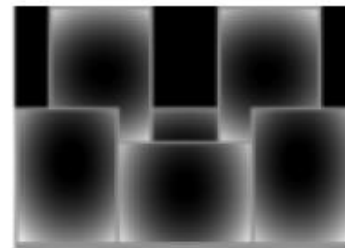
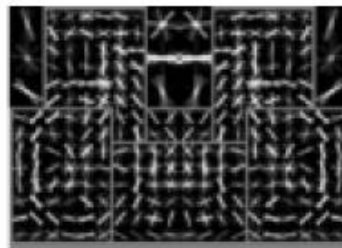
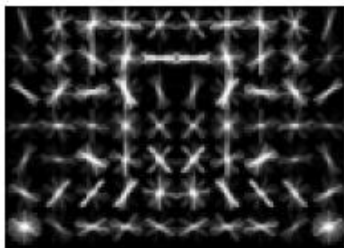
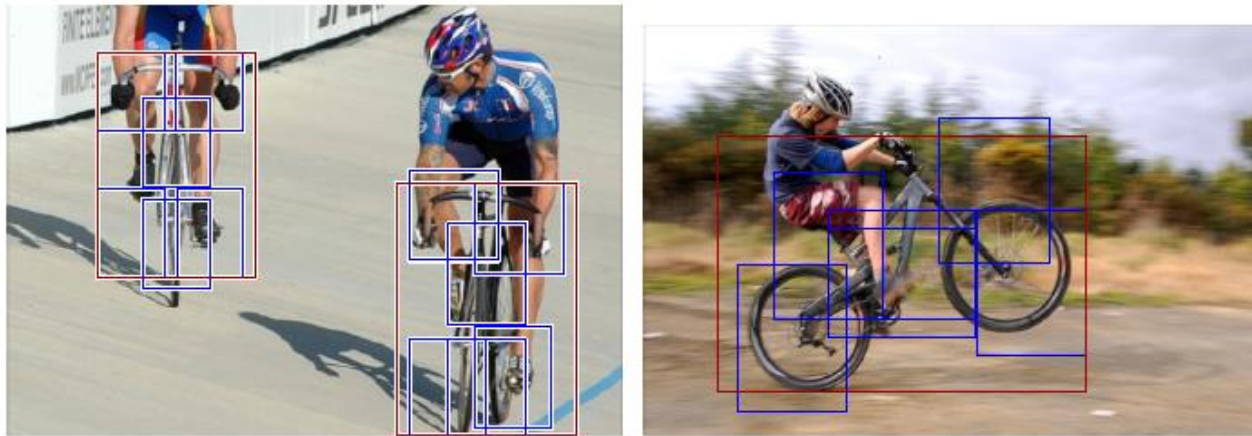


Constellation models





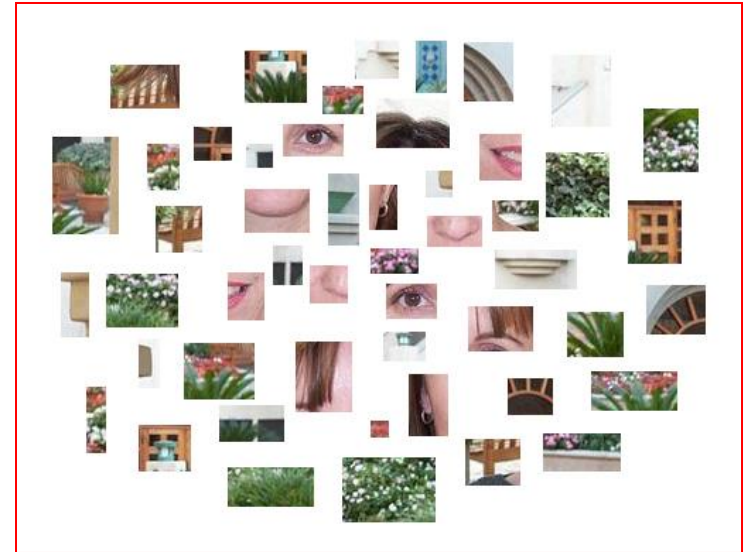
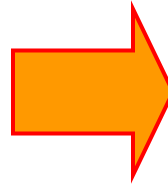
Discriminatively trained part-based models



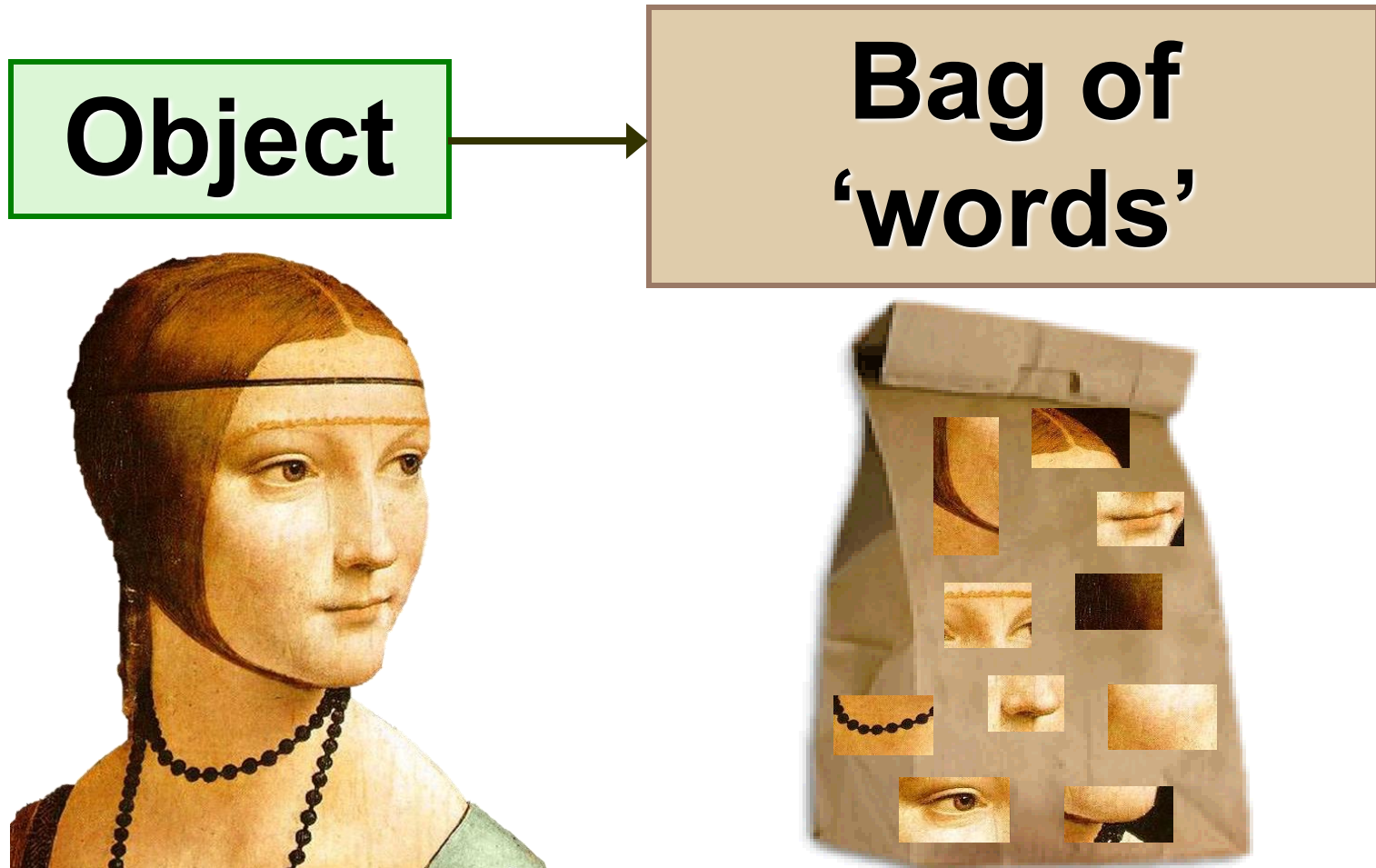
History of ideas in recognition

- 1960s – early 1990s: the geometric era
- 1990s: appearance-based models
- Mid-1990s: sliding window approaches
- Late 1990s: local features
- Early 2000s: parts-and-shape models
- Mid-2000s: bags of features

Bag-of-features models

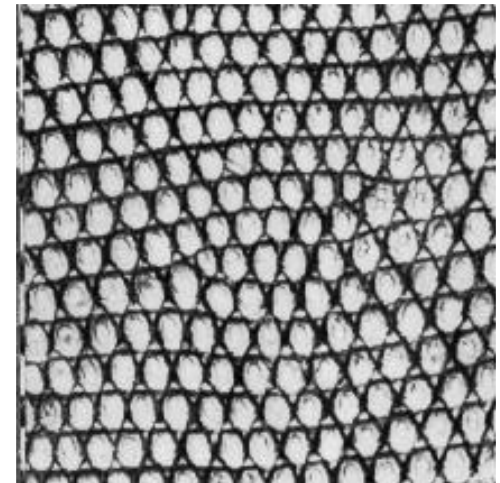
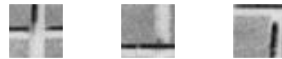
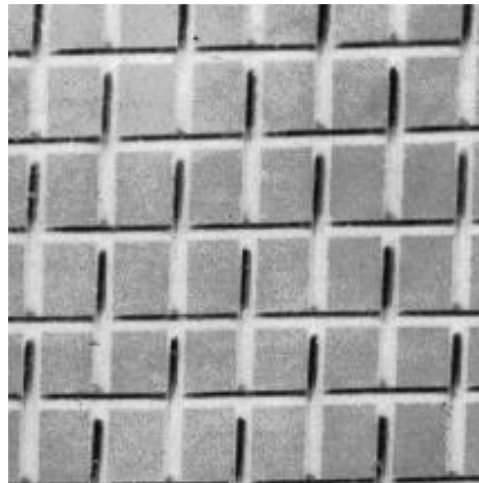
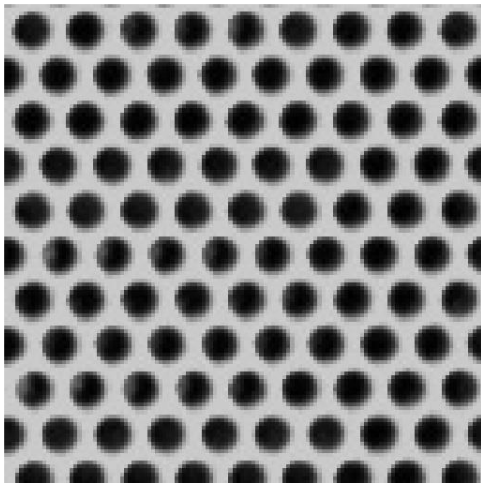


Bag-of-features models

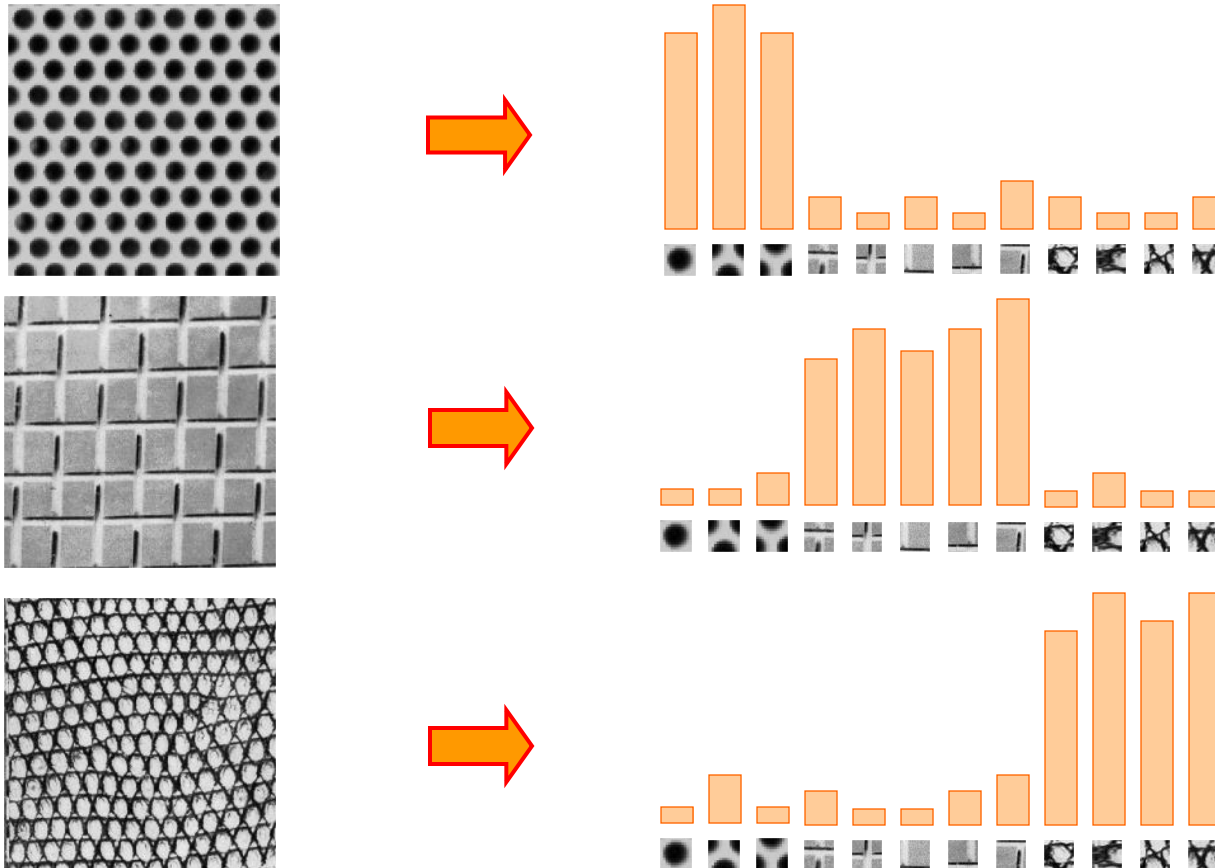


Origin 1: Texture recognition

- Texture is characterized by the **repetition** of **basic elements** or *textons*
- For stochastic textures, the **identity** of the textons is more important than their **spatial** arrangement



Origin 1: Texture recognition



Origin 2: Bag-of-words models

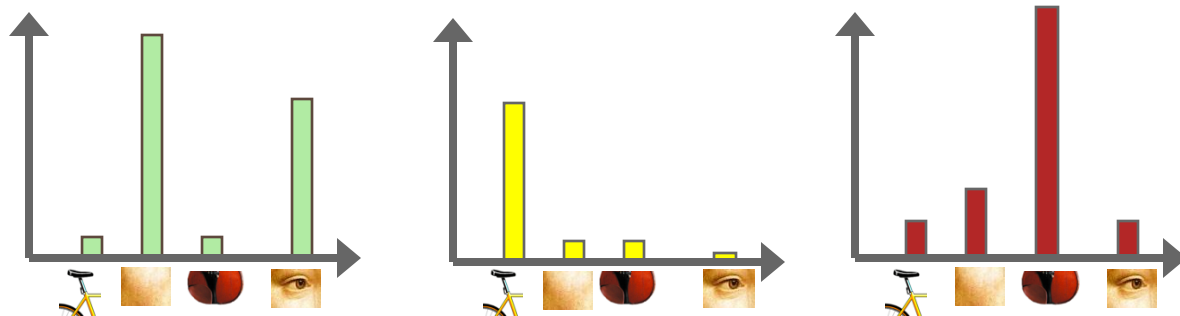
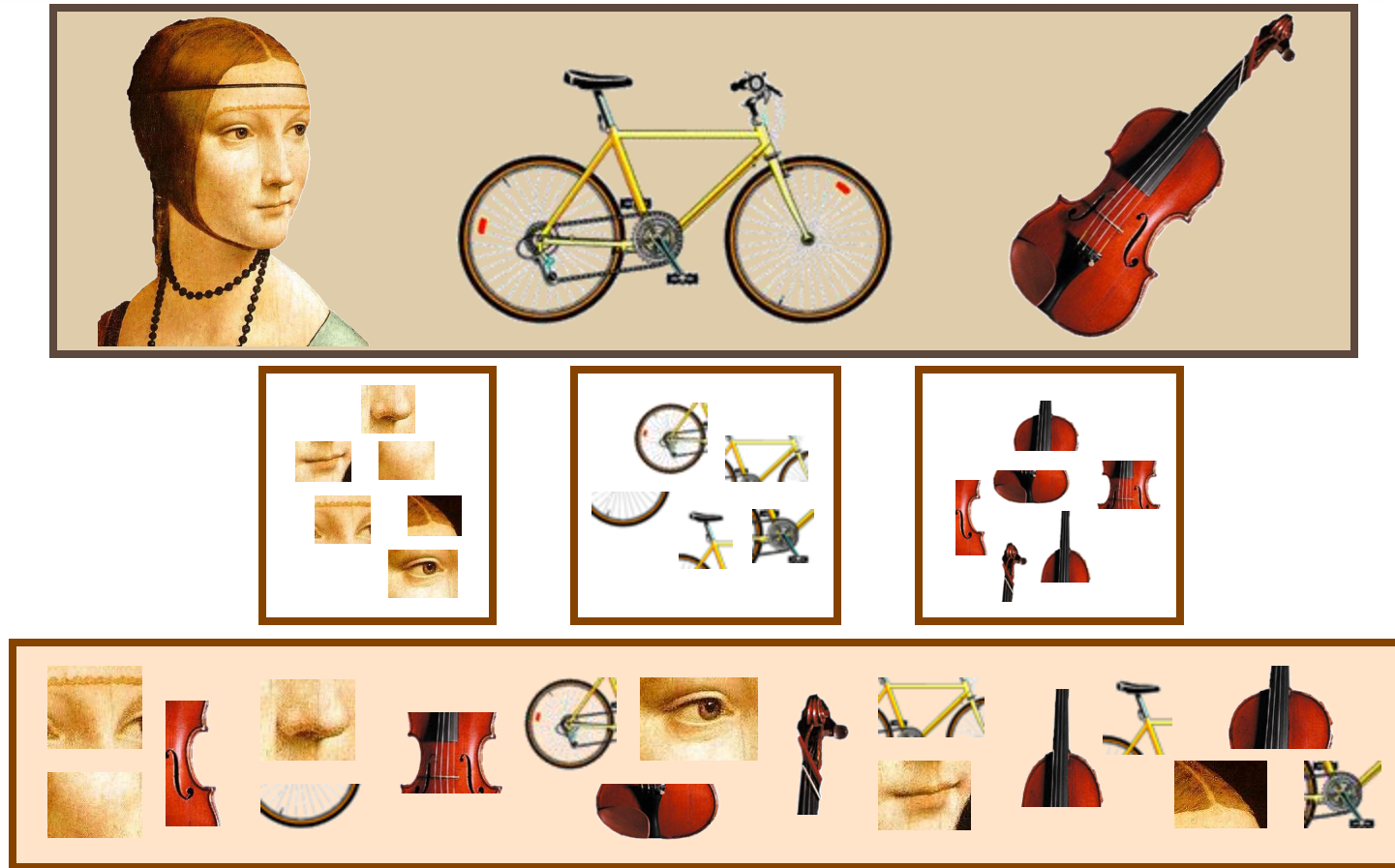
- Orderless document representation: frequencies of words from a dictionary Salton & McGill (1983)



Bag-of-features steps

1. Extract features
2. Learn “visual vocabulary”
3. Quantize features using visual vocabulary
4. Represent images by frequencies of “visual words”

Bag-of-features steps



1. Feature extraction

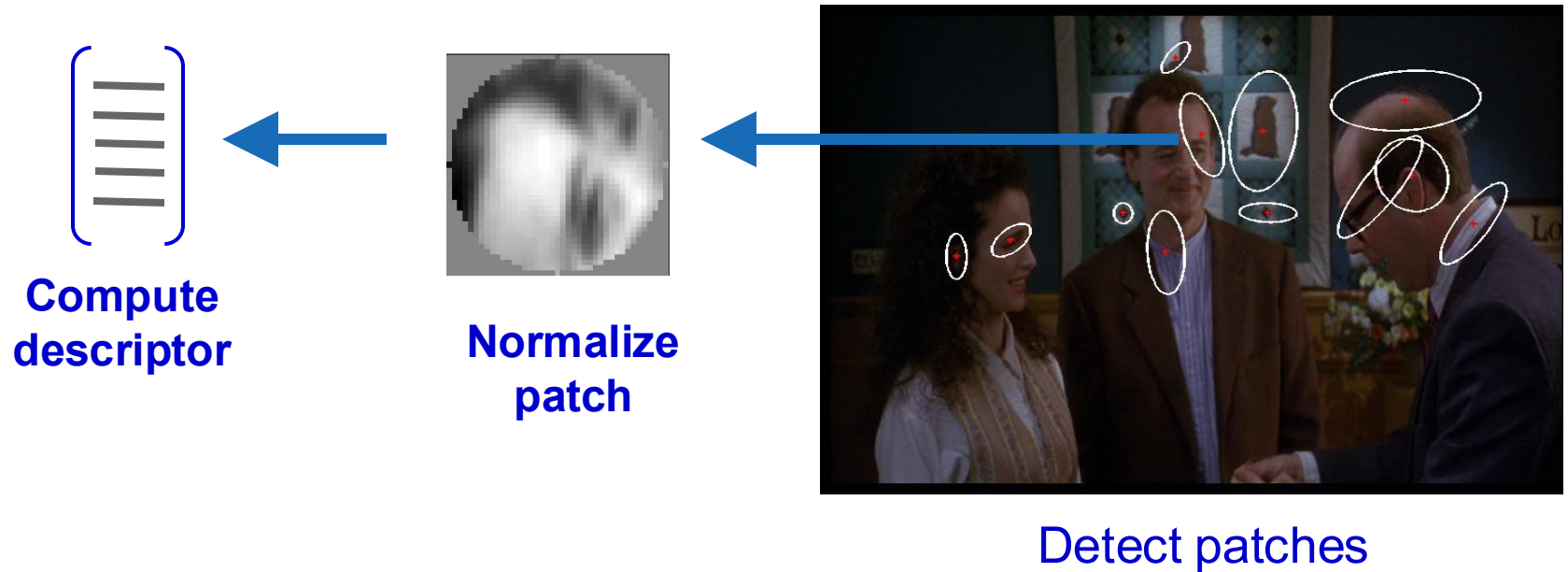
Regular grid



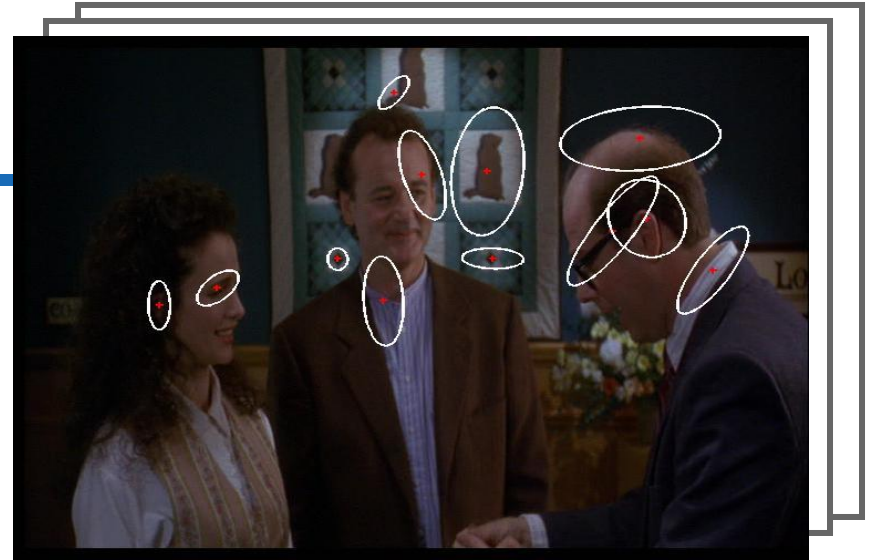
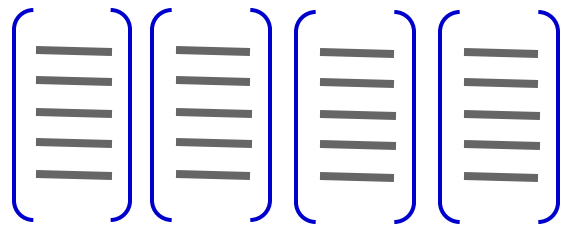
Interest regions



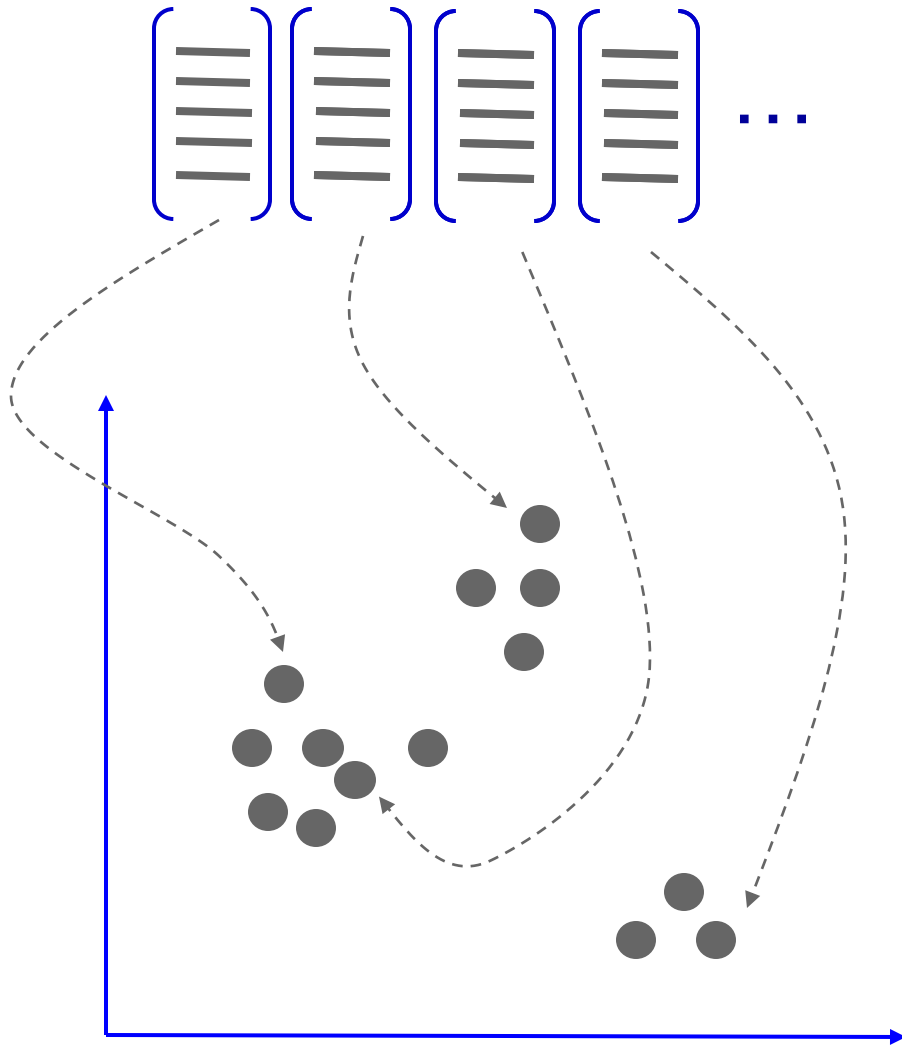
1. Feature extraction



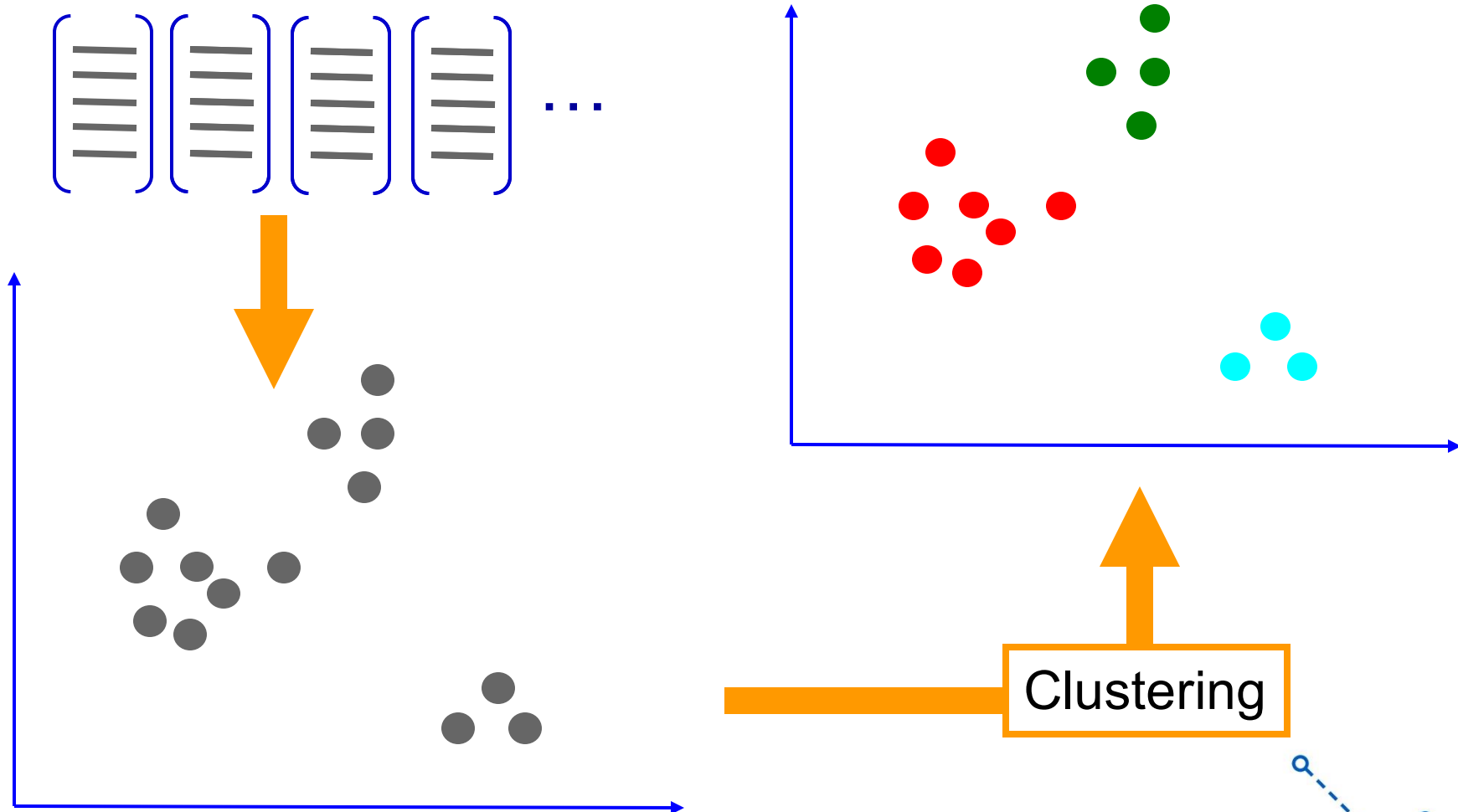
1. Feature extraction



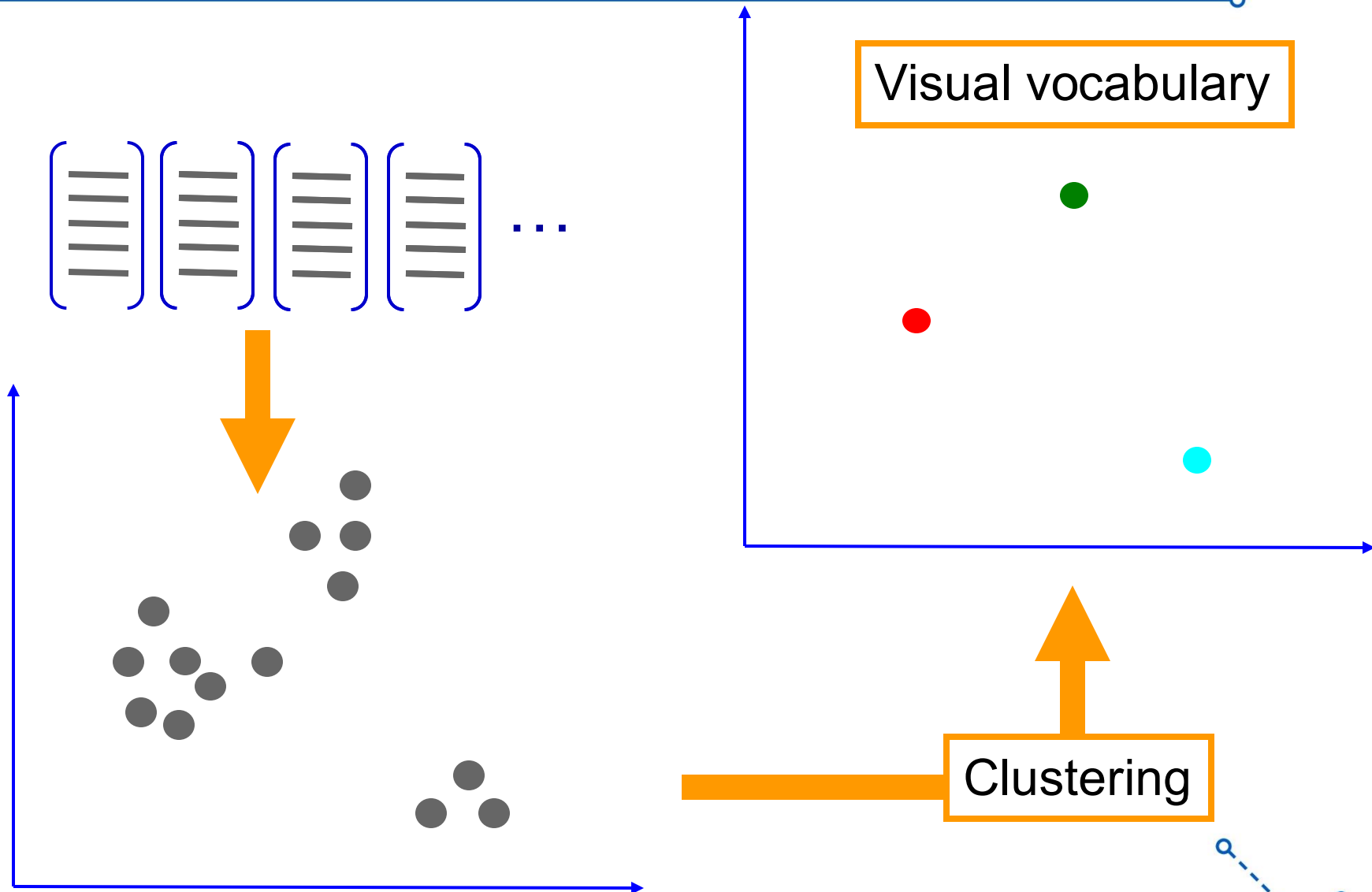
2. Learning the visual vocabulary



2. Learning the visual vocabulary

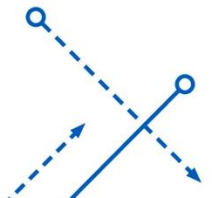


2. Learning the visual vocabulary

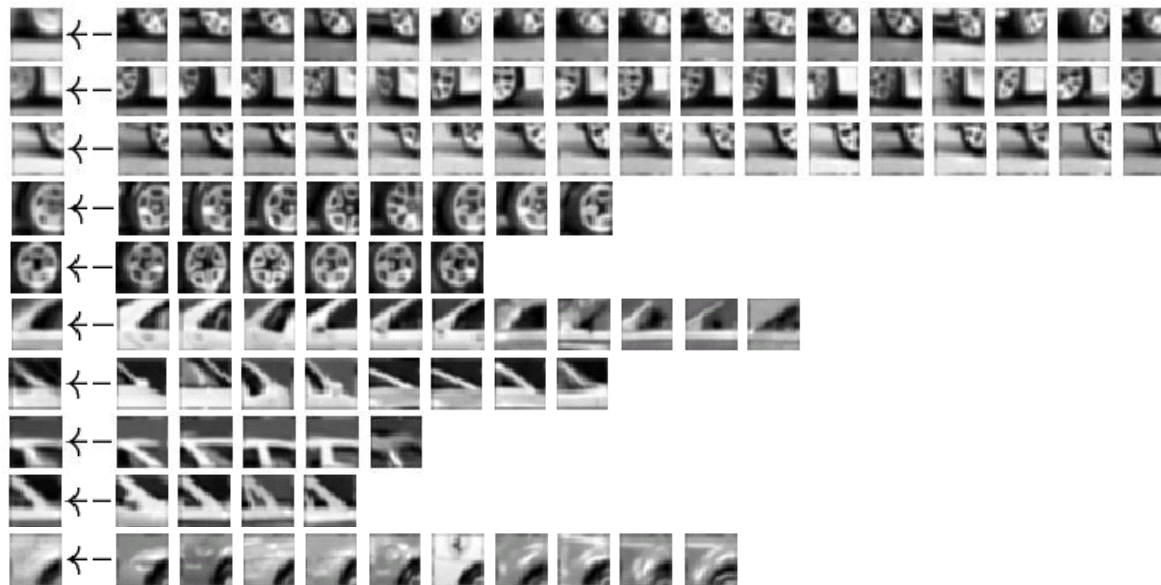
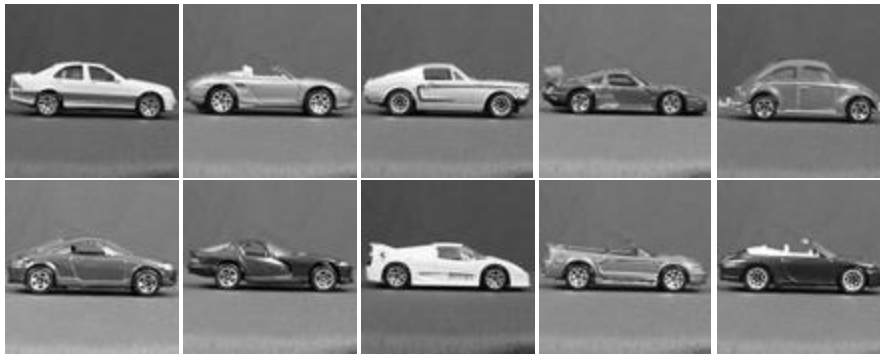


Clustering and vector quantization

- Clustering: learning a visual vocabulary or codebook
 - Unsupervised learning process
 - Each cluster center (e.g., k-means) becomes a codevector
 - Codebook can be learned on a separate training set
 - If the training set is sufficiently representative, the codebook will be “universal”
- The codebook is used for quantizing features
 - A *vector quantizer* takes a feature vector and maps it to the index of the nearest codevector in a codebook
 - Codebook = visual vocabulary
 - Codevector = visual word



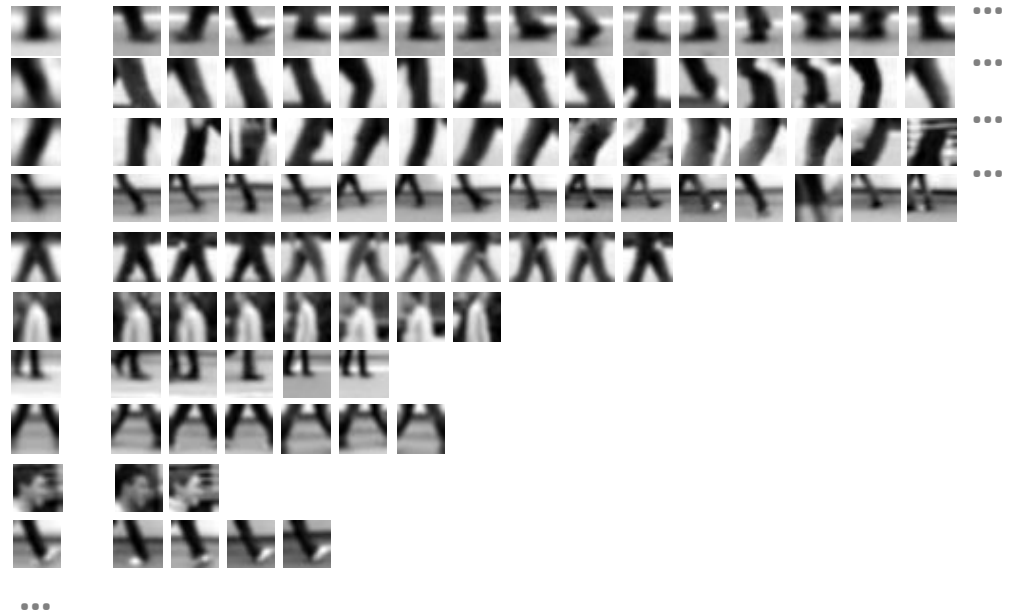
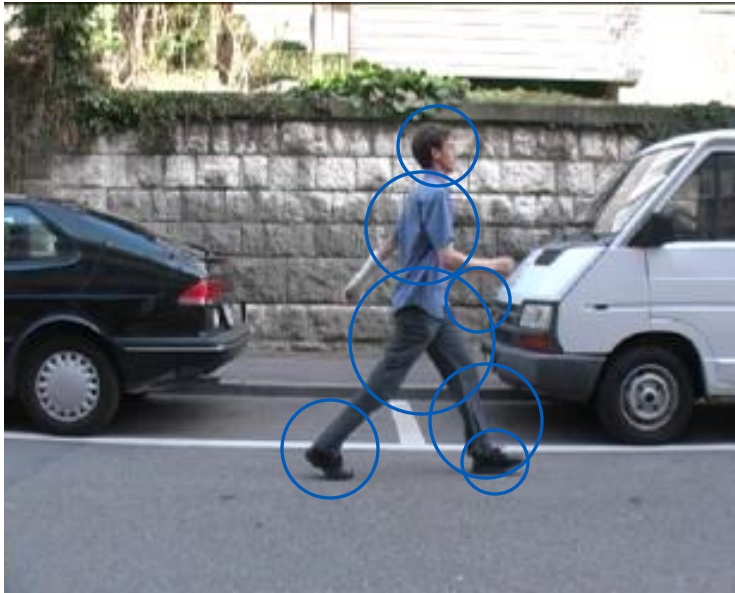
Example Codebook



...

Appearance codebook

Another codebook



Appearance codebook

History of ideas in recognition

- 1960s – early 1990s: the geometric era
- 1990s: appearance-based models
- Mid-1990s: sliding window approaches
- Late 1990s: local features
- Early 2000s: parts-and-shape models
- Mid-2000s: bags of features
- 2010s – Now: deep learning (future lecture)

Conclusion

- Recognition is application-sensitive
- Challenge of Recognition
 - Variation to appearance, scale, articulation, clutter, overlap, and occlusion.
- Bag of words